

- 8 (a) Explain why a body moving with uniform speed in a circle must experience a force towards the centre of the circle.

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..... [2]

- (b) A small ball of mass m is moving in a horizontal circle on the inside surface of a frictionless hemispherical bowl. The normal reaction force N makes an angle θ to the horizontal and is directed along QO , which is the radius of the bowl. The radius of the bowl is 8.0 cm and θ is 22° .

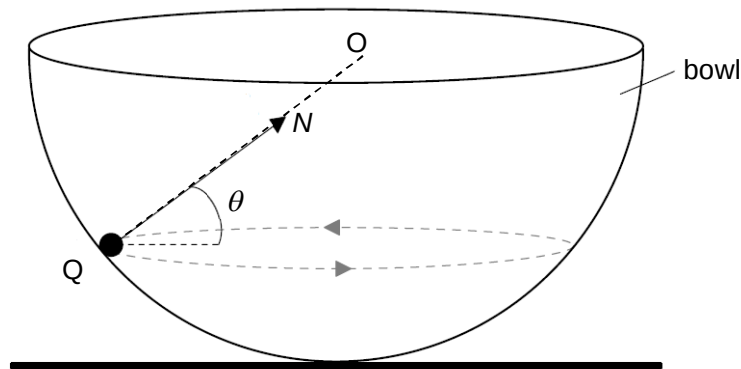


Fig. 8.1

- (i) Determine the speed of the ball.

speed = m s^{-1} [4]

- (ii) Explain whether this ball can move on a horizontal circular path of radius equal to the radius of the bowl.

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 [1]

- (c) A planet has radius R . At a distance h above the surface of the planet, the gravitational field strength is g and the gravitational potential is ϕ .

- (i) Define *gravitational potential* at a point.

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 [2]

- (ii) Show that $\phi = -g(R + h)$.

[2]

(iii) A space probe moves past the planet with speed u as shown in Fig. 8.2.

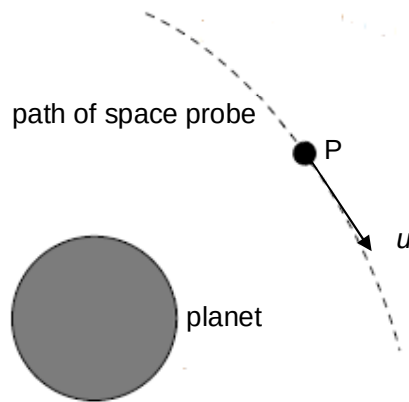


Fig. 8.2

The radius R of the planet is 3.3×10^6 m. A point P, located at 1.4×10^7 m above the surface of the planet, has a gravitational field strength of 0.15 N kg^{-1} .

Determine the minimum value of u such that the space probe at point P is just able to escape from the gravitational attraction of the planet.

$u = \dots\dots\dots \text{ m s}^{-1}$ [3]

- (iv) The variation with distance r (in terms of radius R of the planet) from the centre of the planet of the gravitational potential ϕ is as shown in Fig. 8.3.

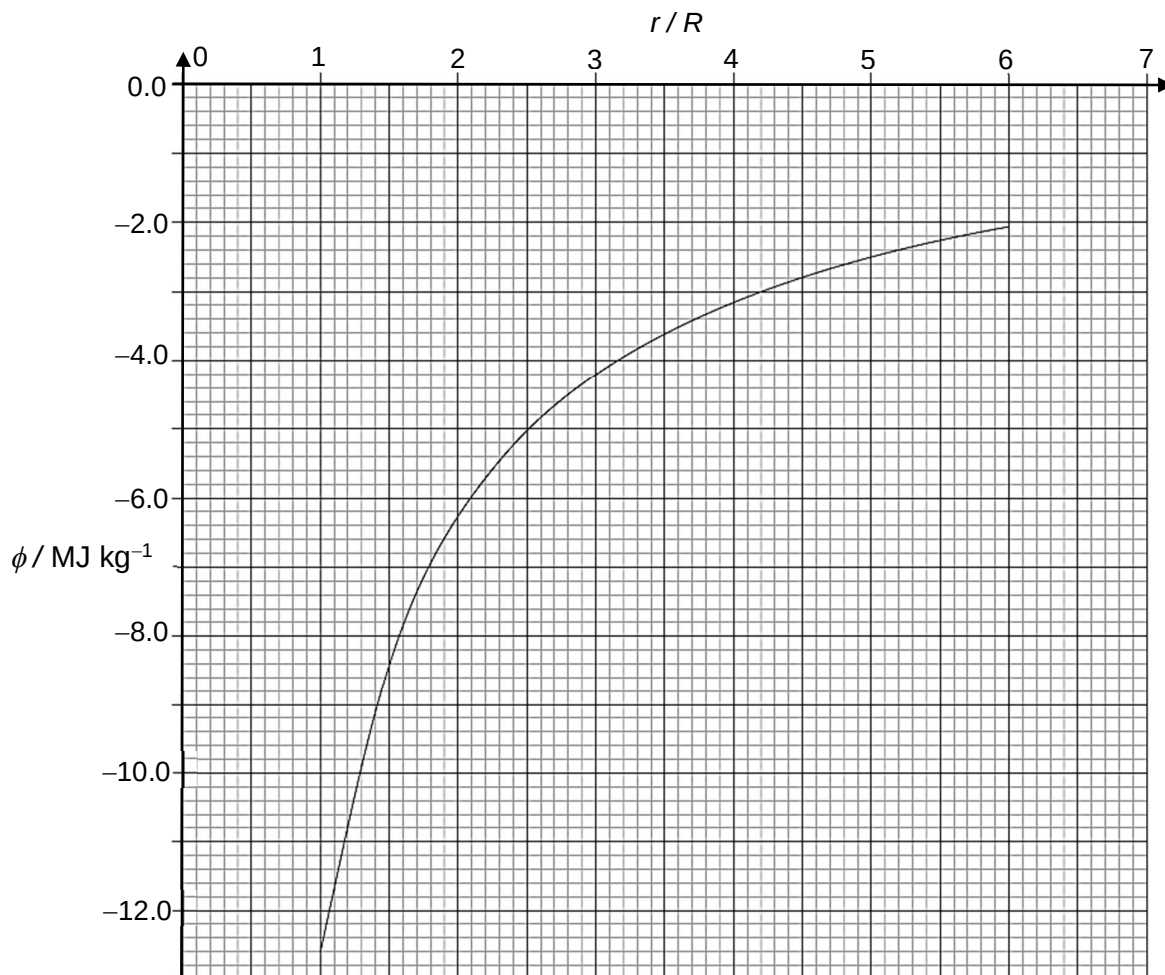


Fig. 8.3

A rocket of mass $1.2 \times 10^4 \text{ kg}$ lifts off from the surface of the planet.

1. Determine the change in gravitational potential energy of the rocket when it reaches a distance $4R$ from the centre of the planet.

change = J [2]

2. Determine the magnitude of the gravitational field strength at a distance $4R$ from the centre of the planet.

magnitude = N kg^{-1} [2]

3. Using the answer from 2., calculate the magnitude of the gravitational field strength at the surface of the planet.

magnitude = N kg^{-1} [2]

9 (a) Define the *volt*